



Exploring the journey of influencers in shaping social media engagement success

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ABSTRACT

This study unfolds nuanced insights into the diverse dimensions dictating the success of social media influencers. Analyzing more than 210,000 social media posts and utilizing the Heuristic-Systematic Model of Information Processing (HSM), this study explores diverse factors, including individual appearance characteristics, depth of persuasive power, and various influencer types. The findings of this study shed light on the distinct impacts of varying influencer archetypes, such as celebrities and micro-celebrities, on user engagement and reveal the nuanced moderating effects of these archetypes on the relationships intertwined with personal attributes, persuasive potency, and influencer success. The proposed model advocates that influencers who leverage more profound, systematic processing strategies, marked by detailed information analysis and conveyance, are poised to experience elevated user engagement compared to counterparts employing heuristic modalities, distinguished by practical mental shortcuts and superficial examinations. This elucidation accentuates the imperative of harmonizing heuristic and systematic methodologies for emerging influencers and brands aspiring to optimize user engagement and efficaciously mold consumer behavior. This paper encapsulates a comprehensive exploration of the dynamic landscapes of influencer marketing via the HSM prism, delivering profound insights and practical ramifications for scholars, marketers, and influencers aiming to navigate and exploit the intricate networks of influential determinants in the ever-evolving digital marketing domain.

1. Introduction

The marketing landscape has been transformed with the dawn of influencer marketing on social media platforms [2,42,62]. Recent surveys indicate that an overwhelming majority of marketers (i.e., exceeding 90 %) are firmly convinced of the utility of influencer marketing [69]. Consequently, this conviction has manifested in an uptick in financial allocation towards influencer marketing strategies [25,26,35,56,83,95]. By providing content on social media platforms, influencers shape their followers' opinions, behaviors, or purchasing decisions [30,34,82]. Moreover, influencers elevate the credibility and trustworthiness of the products or services they endorse [1,39,52,64]. However, identifying those who can deliver tangible results regarding brand engagement and sales uplift remains a formidable challenge [80,81].

Most extant studies have sought answers predominantly concentrating on the sheer volume of an influencer's following as a surrogate marker for influence. A handful of scholars have ventured beyond this limited scope, introducing more sophisticated, multifaceted models for influencer evaluation (Langner et al. [50]; Song [71,85]). For instance,

Rakoczy [66] proposes the concept of micro-influencers, who have significant influence despite having a smaller audience. Although insightful, most of these studies have primarily used surveys to examine persuasion antecedents in isolation from other appearance characteristics [55,63,78]. A burgeoning line of inquiry incorporates computational machine learning techniques to demystify the anatomy of influence [44,91]. Despite these efforts, there is a significant gap in the extant literature on an influencer's capacity to augment user engagement. Therefore, as the first research objective, this study conceptualizes and validates the concept of influencer engagement success in social media.

The appearance of an influencer, embodying aspects like gender and age, plays a pivotal role in relatability and target audience resonance [45,59,75,86,92]. Metrics such as likes and comments gauge persuasive power and suggest which content is worthy of attention and engagement [13,49,51]. These elements coalesce, establishing an influencer's credibility and enhancing their capacity to captivate and persuade their audience [6,55,77]. Thus, as the second research objective, this study explores the effects of influencers' appearance characteristics and

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persuasive powers on influencer engagement success.

Extant literature on influencers argues that influencer type is a good moderator in predicting promotion success in social media [29]. The 'type' of influencer is classified according to follower count into Micro, Macro, Mega, and Celebrity categories [17]. While it is not the only factor to consider, influencer type plays a significant role in predicting an influencer's success for several reasons, including reach and exposure, credibility, trust, and engagement potential [65]. Thus, the third research objective, this study investigates how an influencer's type determines their success.

Drawing on the Heuristic-Systematic Model of Information Processing (HSM) [14,79], this study, finely tuned to Instagram's unique role in influencer marketing, analyzes over 210,000 posts and stories from 2019 to 2023 to explore its research objectives.

2. Theoretical background

2.1. Social media influencers

The economic ramifications of social media influencers are already discernible. According to Statista, the global influencer marketing market value stood at 16.4 billion U.S. dollars as of 2022, having more than doubled since 2019 [21]. These figures underscore the rapid monetization of social influence, compelling scholars and practitioners to better understand the nuances of this emerging field. One of the foundational theories (i.e., consumer's doppelganger effect) underpinning the success of influencers is deeply rooted in social psychology [44]. As conceptualized by Ruvio et al. [67], the consumer's doppelganger effect posits that individuals are inclined to imitate the consumption behaviors and even the physical appearances of their chosen role models. This tendency establishes a unidirectional flow of influence, whereby the consumer draws inspiration from an influencer without necessarily reciprocating that influence (Brown and Taylor, 2019). This principle, parallel to well-known theories such as social proof [5] and mimetic isomorphism [9,32], has immense implications for manufacturers and marketers. Specifically, businesses can amplify their reach and efficacy in target markets by leveraging influencers who exert such unidirectional influence over their followers [38,67,74].

De Veirman et al. [19] define social media influencers as "individuals with extensive networks of followers who are regarded as trusted tastemakers in one or several niches." This definition encapsulates the reach of these influencers, captured through metrics like the number of followers and their capacity to serve as arbiters of taste and credibility within particular domains. Much of the existing literature has embarked on the ambitious endeavor of identifying variables determinative of an influencer's persuasive power [10,20]. The influencer's number of followers, expertise, and physical appearance are among the most frequently cited variables [19,93]. However, it is crucial to note that the influence landscape is complex and multifaceted, extending beyond these conventional indicators. Other influential factors cited in the literature include active minds and social presence [43], expertise in specific subjects [18], the aura of authority that an influencer commands [16], and shared interests between the influencer and their followers [11].

Researchers have applied various methodologies to measure these diverse aspects of influence and persuasion. While some studies have focused on Centrality-based methods [89], which analyze the influencer's position within a network to quantify their impact, others have turned to well-established algorithms like PageRank [7]. More contemporary approaches have adopted specialized algorithms like TwitterRank [91] that are tailored to understand influence in the specific context of social media platforms. Each method sheds light on different facets of what is increasingly understood as a highly complex construct of persuasion power. While existing research has made valuable contributions to understanding the architecture of social media influence, it is important to acknowledge each methodology's and

conceptual framework's limitations. For instance, most of these studies have primarily explored these variables in isolation or, at best, in limited combinations. There is a glaring gap in the literature regarding amalgamating these individual variables into a composite, holistic framework that could robustly predict an influencer's capacity to impact consumer choice and behavior. By offering an exhaustive review of these theoretical underpinnings, this study provides a foundational basis upon which this study will further delve into the intricacies of influencer success, thereby contributing to advancing theoretical constructs and practical applications in digital marketing.

2.2. Heuristic-Systematic Model of Information Processing (HSM)

The Heuristic-Systematic Model of Information Processing (HSM) [14,79] is a widely recognized psychological theory developed to elucidate how individuals manage information and form attitudes, thereby enabling an understanding of persuasion. The model postulates two parallel modes of information processing: heuristic and systematic. According to this model, heuristic processing involves the utilization of mental shortcuts ('heuristics') to form judgments and make decisions quickly and with minimal cognitive effort. Heuristics are rules of thumb that individuals employ to simplify complex cognitive tasks, thus allowing them to navigate the constant stream of incoming information more efficiently. This processing mode does not delve into detailed analysis but relies on superficial cues and existing knowledge structures. For instance, a common heuristic is the "expertise heuristic," where individuals may attribute credibility to a communicator based on perceived expertise, often without critically examining the content of the communication. Contrasting with heuristic processing, systematic processing is a more deliberate, effortful, and thorough mode of evaluating information. It involves scrutinizing available data, considering various perspectives, and weighing evidence before forming a judgment or making a decision [14,79]. This detailed analysis allows individuals to develop more nuanced and informed attitudes and opinions, usually resulting in more stable attitude changes.

The HSM proposes that these two processing modes can coexist and interact within the persuasion process. Individuals may employ both modes simultaneously or oscillate between them based on motivation, ability, and the availability of cognitive resources. The amount and nature of cognitive processing occurring in a given situation depends on individual and contextual factors, including the topic's relevance, distractions, and the individual's need for cognition. The HSM provides a comprehensive framework for understanding how consumers process influencer-generated content in social media and influencer marketing. The heuristic processing mode may dominate when consumers rapidly scroll through their social media feeds, relying on heuristics such as influencer credibility or the number of likes to form quick judgments about products or brands [76]. In contrast, when consumers encounter content related to a high-involvement product or a topic of significant personal relevance, they may engage in more systematic processing, critically evaluating the information presented by the influencer and considering various product attributes and alternatives before forming an attitude or making a purchase decision [47]. By recognizing the dual processing paths within the HSM, we can gain deeper insights into consumer behavior and devise more effective communication and persuasion strategies tailored to consumer needs and contexts.

3. Research questions and hypotheses

Drawing from the rich tapestry of existing scholarly work, our research embeds its foundation within the HSM [14]. Recognizing the layered complexities of modern digital interactions, this study endeavors to embellish prior research. Specifically, we aim to holistically consider the symbiotic influence of personal attributes (often referred to as heuristics in academic circles) and the potency of persuasive power (or systematic influence) on the intricate dynamics of user engagement.

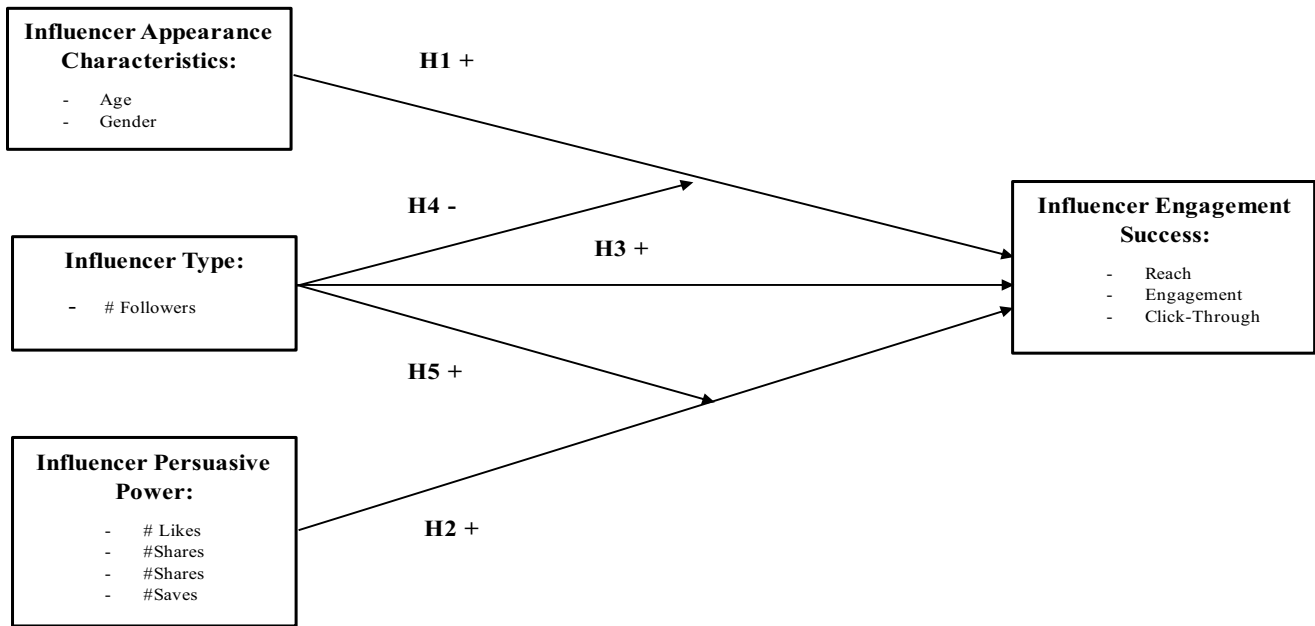


Fig. 1. Research model.

Fig. 1 comprehensively conceptualizes our proposed model to visualize this multi-pronged research paradigm.

In the ever-evolving landscape of digital interactions, understanding the nuances and implications of personal attributes on influencers' success is paramount. Historical research has shed light on the fact that perceptions surrounding an advertised product are intricately linked to the gender and physical allure of models in visual advertisements [40]. With the advent and proliferation of social media, these effects have also found their resonance within the domain of social media influencers. For instance, a study by Gilbert et al. [28] suggests that while men are often seen to command larger follower bases, women influencers have an uncanny ability to spur higher levels of engagement, demonstrated by tangible metrics such as reach and engagement. In the context of social media, it is argued that younger-looking females are more successful in engaging with their followers [53,75,80,92]. The HSM provides valuable insights into the prominence of younger-looking females on social media, particularly in their ability to attract more attention and engage more effectively with their followers. Within the HSM framework, the heuristic route is a fundamental aspect. This route is characterized by the human tendency to make rapid judgments and decisions using cognitive shortcuts or heuristics, often due to the sheer volume of information on social media platforms [76]. Individuals often rely on these mental shortcuts when browsing their feeds to quickly assess and engage with content [72]. Among the most influential heuristics in this context are attractiveness and familiarity [57]. Younger-looking females have an advantage in appealing to these heuristics because they typically align with societal beauty standards associated with youth [46]. Elements such as a youthful appearance serve as cues that trigger the heuristic of attractiveness. As a result, viewers are more likely to perceive these individuals as physically attractive, garnering their attention.

Additionally, youthfulness contributes to the heuristic of familiarity, as younger individuals often resemble the viewers' peers or aspirational figures. Consequently, this understanding makes it easier for social media users to relate to these individuals, considering them as relatable or role models [4,12,41]. Moreover, youthfulness can also be seen as novel, signifying that these individuals are in tune with the latest trends and cultural shifts. These further captures viewers' attention and encourages them to follow and engage with their content. In essence, younger-looking females benefit from the heuristic route's cognitive shortcuts, initially leading to higher visibility and engagement on social

media platforms. Hence, it is hypothesized that:

H1: Social media influencer's individual appearance characteristics are positively associated with his /her perceived success in social media engagement such that young females are more likely to be successful in engaging with their followers on a social media platform.

Drawing inspiration from the profound insights offered by Chen and Chaiken [14], persuasion emerges as the bedrock principles and mechanisms through which people's attitudes, beliefs, and behaviors transform, adjust, or demonstrate resilience under external influences. In this intricate dance of digital dynamics, content steeped in persuasive eloquence can be a potent catalyst, invigorating user engagement [70, 90]. Conversely, posts that are purely informative, bereft of persuasive undertones, may inadvertently dampen user engagement. The concept of persuasive power in the context of social media influencers aligns closely with the systematic processing route elucidated by the HSM. Within this framework, metrics such as number of likes, shares, comments, and saves on influencer content can exemplify systematic engagement by users. When viewers 'like' a post, they engage in a deliberate evaluation, expressing their approval and appreciation for the content. This action signifies a systematic processing of the material, indicating that it has persuaded or captivated them [61]. Similarly, sharing content extends beyond interaction, representing an endorsement. Users systematically decide to promote content they find persuasive or valuable to their social networks, actively participating in disseminating it [58,94].

Furthermore, comments on influencer posts often showcase a deeper level of systematic engagement. Thoughtful responses demonstrate that viewers have engaged substantively with the content by sharing their thoughts, asking questions, or offering feedback [33,36,87]. Saving content for future reference is another systematic action, signaling that viewers deem the content valuable enough to revisit. The choice to save material highlights the persuasive impact of the content, as individuals systematically opt to engage with it over an extended period. Influencers who consistently accumulate more likes, shares, comments, and saves typically offer content that systematically resonates with their audience [15]. This deep connection between influencers and followers leads to several favorable outcomes.

Content with more likes and shares often enjoys an extended reach, echoing the principles of the systematic processing route in HSM. Influencers who thoughtfully share content, contributing to its success by expanding their audience [84,88]. Moreover, comments on influencer posts foster a sense of community and initiate discussions, intensifying engagement [26,60]. This systemic interaction enriches the bond between the influencer and their audience. Additionally, content that persuades viewers to save it for later tends to result in higher click-through rates when the saved content is revisited. This underlines the persuasive power of systematic engagement, as individuals are more inclined to take further actions, such as clicking on links, making purchases, or engaging more deeply with the influencer's content beyond the initial encounter. Having all said, persuasive power, as evident in metrics like likes, shares, comments, and saves, is intrinsically tied to the systematic processing route detailed in the HSM Model. This systematic engagement profoundly influences the success of social media influencers and their content, translating into amplified reach, heightened engagement, and increased click-through rates. It emphasizes the significant impact of a systematic content evaluation on social media influence. Hence, it is hypothesized that:

H2: The depth of persuasive power infused within a social media influencer is positively associated with its success, such that influencers with higher persuasive power are more likely to be successful in engaging with their followers on a social media platform.

The number of followers is a valuable proxy for measuring persuasiveness in social media as it reflects an influencer's ability to capture the audience's attention [93]. In this regard, HSM sheds light on the significance of the number of followers as a heuristic metric for measuring engagement in social media. Social media users often rely on heuristics or cues to make judgments. Users tend to perceive accounts with a large following as more credible, influential, and reliable content sources, leading to higher engagement [73,93]. Beyond mere popularity, the content of a social media influencer with a high follower count due to the network effect may attract more users' attention, leading to higher chances of user engagement in social media [19,80,93]. Therefore, it is hypothesized that:

H3: Influencer type is positively associated with the success of influencers in generating user engagement on social media.

With significant followers, the influencer's content tends to be perceived as more trustworthy ([3], p. 220). Therefore, as an influencer's number of followers increases, it is anticipated that the influence of personal appearance characteristics, such as age and gender, on engagement success will diminish. As an influencer's popularity and reach expands, their ability to connect with a diverse and broad audience becomes a pivotal factor in their success in engaging social media users [93]. Moreover, utilizing the HSM in the context of social media influence provides an interesting lens through which to examine how follower count, as a heuristic cue, might mitigate the impacts of demographic variables such as age and gender in shaping user engagement. Heuristic processing, characterized by using mental shortcuts or cues (e.g., high follower numbers) for decision-making, might prompt users to engage with content. When an influencer has more followers, users might engage based on the heuristic that "many followers equal quality or entertaining content," potentially sidelining any biases or preferences related to the influencer's age or gender. In this scenario, despite age and gender traditionally being factors in engagement, a significant follower count can act as a powerful heuristic that diminishes the weight of these variables, facilitating engagement based on perceived popularity and credibility. Hence, it is hypothesized that:

H4: Influencer type negatively moderates the impact of personal properties on influencers' success in users' engagement.

Extrapolating from existing narratives, celebrity influencers, bolstered by their expansive recognition and clout, are often perceived to possess an augmented capacity to sway audiences [8]. When content posted by a social media influencer receives more likes, shares, comments, and saves, the presence of a large follower base amplifies the significance of these engagement factors in determining the influencer's success in engaging with users. This amplification is primarily due to the social proof and credibility associated with many followers [96]. As followers perceive the influencer as a trusted content source, they are more inclined to engage with posts that have already garnered considerable likes, shares, comments, and saves. This compounding effect fosters community and trust, increasing interaction and engagement.

Furthermore, influencers with more followers tend to have significantly higher reach and visibility of their content. As persuasiveness metrics increase, the content resonates with the influencer's immediate followers and extends to a broader and more diverse audience. Social media algorithms also favor content with high engagement, promoting it to a wider user base [22]. Consequently, the synergy between a high follower count and heightened persuasiveness metrics propels an influencer's content to the forefront, increasing the chances of engagement. Hence, it is hypothesized that:

H5: Influencer type positively moderates the impact of the persuasive power of influencers on influencers' success in users' engagement.

4. Data collection

This study collected its data from Instagram using publicly available Instagram APIs. We have exclusively collected data from Instagram to focus our study, acknowledging that it may restrict our understanding of platform-specific biases without comparison to other platforms. We supplemented the data by writing a Python script, using two libraries of "BeautifulSoup" from "bs4" and "requests" that received @username as input and retrieved the number of account-related data. We also used a similar method to prepare another Python script that received social media post URLs to extract post-related information. We have collected data in two categories: account-related data, such as the number of followers, gender, and age of the influencer, and post-related data, such as the number of likes, comments, saves, and clicks. The final filter employed was for posts and stories containing links shared by the influencer, enabling us to gather click-through data and contribute to ensuring influencer engagement success. It is worth noting that all the constructs measured in the research model are formative. Table 1 provides detailed information about the constructs and their associated indicators extracted from Instagram.

The final dataset included 210,000 social media posts and stories by influencers on Instagram in the USA over the past four years (i.e., 2019–2023) that have received high engagement among social media users. Selecting the influencers followed the recommendation provided by HypeAuditor [37]. Table 2 categorizes the influencers [25,68].

Finally, the selection of age thresholds at 35 and 55 years was

Table 1
Constructs and their associated indicators.

Constructs	Indicators
Influencer's Appearance characteristics	• Age
Persuasive Power	• Gender
	• Number of Likes
	• Number of Comments on the Post/Story
	• Number of Shares
	• Number of Saves
Influencer Engagement Success	• Reach
	• Engagement
	• Number of Click-Throughs
Influencer Type	• Number of Followers

Table 2
Influencer type.

Influencer Type	Number of Followers
Nano-Influencer	1 K to 10 K
Micro-Influencer	10 K to 50 K
Macro-Influencer	50k to 500K
Mega-Influencer	500 K to 1M
Celebrities	Over 1 M

informed by demographic segmentation principles commonly utilized in marketing research, as outlined by Kotler and Keller [48]. This segmentation enables a nuanced understanding of consumer behavior across distinct life stages, aligning with the focus of our study on influencer marketing and enhancing the relevance of our findings.

5. Data analyses and results

5.1. Descriptive statistics

Before any analysis, two main modifications were conducted on the collected indicators of the constructs in the research model of this study. First, age and gender were transformed into categorical variables. Several reasons for this transformation include controlling the effects of the non-linearity of variables, increasing the interpretability of the results, and controlling the heterogeneity of indicators. Next, this study used logarithms instead of the actual numbers to prevent the effects of significant big numbers on the rest of the indicators to have a normalized distribution. This technique enables a more refined analysis of how various engagement factors impact influencer success and delivers more accurate insights into the dynamics of user interactions on social media platforms. Following the above ramifications, Tables 3–5 present the descriptive statistics for the constructs' indicators in this study's research model.

5.2. Validating measurement model

All the constructs in the research model of this study are formative. Therefore, before testing the hypotheses, to validate the measurement model and the constructs in the research model, this study followed Hair et al. [31] recommendations. According to Hair et al. [31], to validate formative constructs, scholars should conduct two series of analyses: (i) collinearity analysis and (ii) indicator significance analysis. As the first step, we checked the collinearity among indicators. As shown in Table 6, this study and its collected data do not suffer from any collinearity issues among the constructs' indicators in this study's research model.

As the next step in collinearity analysis, we also looked at the collinearity among the formative latent constructs. As in the previous step, this analysis resulted in similar findings. Therefore, it could be concluded that this study does not suffer from any collinearity issues. Results are depicted in Table 7.

As the next step, this study followed Gefen and Straub's [27] recommendations to check the statistical significance of indicators shaping the formative constructs. In this regard, T-values and outer weights were analyzed. As shown in Table 8, the results yielded that all the indicators satisfy the requirements for validating formative constructs.

As shown in Table 8, gender is the most prominent factor shaping influencers' appearance characteristics. Moreover, among the indicators

Table 3
Descriptive statistics of age.

Indicator	Age 1	Age 2	Age 3
Age	105,657	89,176	15,342

Total Number of Posts / Stories Collected = 210,175.
Age (1 (i.e., Young) <=35; 2 (i.e., Middle age <=55); 3 (i.e., Old > 55)).

Table 4
Descriptive statistics of gender.

Indicators	Gender 1	Gender 2
Gender	128,596	81,579

Total Number of Posts / Stories Collected = 210,175.
Gender (1 representing female, and 2 representing male).

Table 5
Descriptive statistics of the rest of indicators of constructs in the research model.

Indicators	Mean	Median	Standard Deviation	Min	Max
Log (Number of Likes)	4.11	4.07	1.26	1.12	7.87
Log (Number of Comments)	4.61	4.50	1.27	1.07	7.63
Log (Number of Shares)	4.52	4.11	1.24	0.65	6.12
Log (Number of Saves)	4.27	4.12	1.17	0.12	4.85
Log (Reach)	4.35	4.13	1.38	1.19	8.12
Log (Engagement)	4.21	4.07	1.02	1.12	6.25
Log (Number of Click-Throughs)	2.57	2.12	1.22	0.08	4.85
Log (Number of Followers)	5.7	5.66	1.12	1.01	7.60

Total Number of Posts / Stories Collected = 210,175.

shaping persuasive power, the number of likes is the most prominent factor. Engagement is also the most significant indicator shaping influencer engagement success.

Furthermore, we checked the multicollinearity, following the extant recommendations [24,54]. As shown in Table 9, since all VIF levels are below 3, we can conclude that this study does not suffer from the issue of multicollinearity.

As the last step in validating the research model, we analyzed its discriminant validity. This analysis revealed that this study is prone to any discriminant validity-related issues.

5.3. Structural model analysis and hypotheses testing

To test the hypotheses associated with this study's research model, we employed a structural equation modeling technique using SmartPLS 3.3.2. Accordingly, following extant literature recommendations, a bootstrapping method was used, assuming the data is normally distributed, with 10,000 re-samplings [23]. Fig. 2 depicts the results.

As shown above, not all the hypotheses of this study were supported. To be more specific, interestingly as opposed to what has been proposed in H1, it was found that individual appearance characteristics of an influencer do not associate with the influencer's engagement success. H2 was supported, revealing that posts that receive more likes, shares, comments, and saves bring more chances of engagement success, encouraging social media users to follow the links. Same with H3, as proposed in this hypothesis, it was found that the influencer type positively correlates with an influencer's engagement success. Therefore, it could be concluded that the number of followers, as the proxy to measure influencer type, potentially is a good indicator of a social media post's success in bringing more engagement.

Moreover, in line with the proposed H4, it was found that Influencer type negatively moderates the impact of influencers' appearance characteristics on engagement success. The higher the number of followers an influencer has, the less social media users care about the appearance characteristic of such an influencer. Fig. 3 illustrates this interaction.

Finally, H5 was also supported, reflecting that the higher the number of followers following, the higher the persuasiveness of a social media post, and the higher the chances of engagement success of such a social media post. Fig. 4 illustrates the results of this interaction.

Additionally, as shown in Fig. 2, the r-squared for this study showed a value of 0.629, exceeding the requirement for the accepted r-squared

Table 6

Formative variable indicators correlations.

	Age	Gender	Log (Number of Likes)	Log (Number of Shares)	Log (Number of Shares)	Log (Number of Savve)	Log (Number of Reach)	Log (Number of Engagement)	Log (Number of Click-Throughs)	Log (Number of Followers)
Age	1.00									
Gender	0.42	1.00								
Log (Number of Likes)	0.21	0.26	1.00							
Log (Number of Comments)	0.28	0.21	0.20	1.00						
Log (Number of Shares)	0.20	0.16	0.59	0.58	1.00					
Log (Number of Saves)	0.32	0.14	0.54	0.54	0.55	1.00				
Log (Reach)	0.26	0.11	0.57	0.59	0.58	0.55	1.00			
Log (Engagement)	0.23	0.05	0.54	0.48	0.48	0.52	0.52	1.00		
Log (Number of Click-Throughs)	0.21	0.01	0.43	0.10	0.08	0.13	0.61	0.56	1.00	
Log (Number of Followers)	0.07	0.11	0.52	0.58	0.48	0.38	0.49	0.51	0.15	1.00

Table 7

Latent formative variable correlations.

	Influencer's Appearance Characteristics	Persuasive Power	Influencer Type	Influencer Engagement Success
Influencer's Appearance Characteristics	1			
Persuasive Power	0.29	1		
Influencer Type	0.22	0.58	1	
Influencer Engagement Success	0.19	0.59	0.55	1

Table 8

Formative construct validation.

Relationship	Weight	T-Value	Keep/Drop
Age → Influencer's Appearance Characteristics	0.512	86.12	Keep
Gender → Influencer's Appearance Characteristics	0.661	110.54	Keep
Log (Number of Likes) → Persuasive Power	0.720	130.15	Keep
Log (Number of Comments) → Persuasive Power	0.141	24.85	Keep
Log (Number of Shares) → Persuasive Power	0.195	32.54	Keep
Log (Number of Saves) → Persuasive Power	0.112	19.16	Keep
Log (Reach) → Influencer Engagement Success	0.532	89.27	Keep
Log (Engagement) → Influencer Engagement Success	0.570	91.50	Keep
Log (Number of Click-Throughs) → Influencer Engagement Success	0.133	22.21	Keep

Table 9

VIF.

	Influencer Engagement Success
Influencer's Appearance Characteristics	1.2
Persuasive Power	2.6
Influencer Type	2.1

range. This implies that the proposed research model and the collected dataset for this study were significantly able to capture the factors shaping and affecting the concept of influencer engagement success.

5.4. Goodness of fit analysis

For the final assessment, we checked the goodness of fit of collected data with the research model of this study. This exploration indicated an SRMR value of 0.015 and an NFI value of 0.976. According to the extant literature's recommendation, since these values fall within the accepted

range of a robust model fit (i.e., 0.08 or lower for SRMR and 0.95 or higher for NFI), we can conclude that the collected data aligns with this study's research model and the findings of this study are prone to any goodness of fit related issues.

6. Discussion

In this analysis, the Heuristic-Systematic Model (HSM) served as a foundational framework for interpreting the intricate interplay between individual influencer appearance characteristics, persuasive power, and the types of influencers related to social media engagement success. The model hinges on two primary modes of information processing: heuristic, which involves quick, mental shortcuts, and systematic, entailing a more comprehensive, analytical approach to decision-making. The results corroborate and challenge various dimensions of this model, elucidating the nuanced dynamics within the digital influence landscape. Unlike our initial first hypothesis, an influencer's appearance characteristics, as a heuristic cue, do not significantly affect influencers' perceived success. Therefore, Followers, primarily those less motivated or able to engage in-depth processing, do not rely on these cues to determine the trustworthiness and attractiveness of the influencer's content. The disconfirmation of H1 suggests that individual appearance characteristics are not crucial heuristic cues that affect the perceived success of influencers in social media engagement.

Our second hypothesis findings align with the HSM theory's systematic processing component. When followers scrutinize the content intensely, assessing the persuasive depth and strength of the messages, the influencers succeed in their posts. Integrating persuasive elements in an influencer's post encourages a meticulous analysis of the content, impacting its overall success through increased user engagement. The support for the proposed H2 emphasizes the role of systematic processing in the HSM theory. The observed success in posts with more likes, shares, comments, and saves highlights the importance of persuasive depth in content. Influencers' followers engaging in systematic processing meticulously analyze these elements, forming

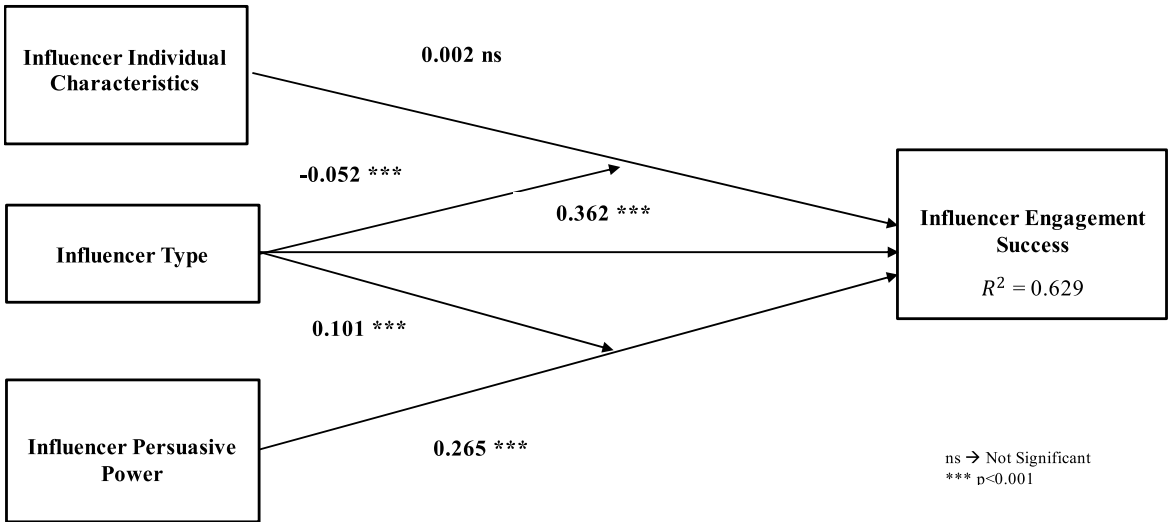


Fig. 2. Results.

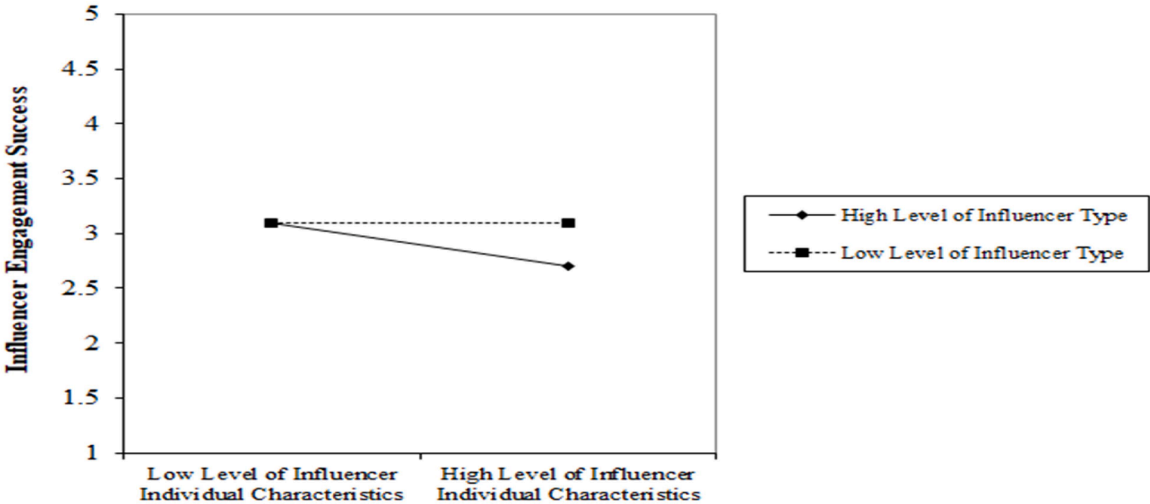


Fig. 3. Moderation effect of influencer type on the relationship between influencer individual characteristics and influencer engagement success.

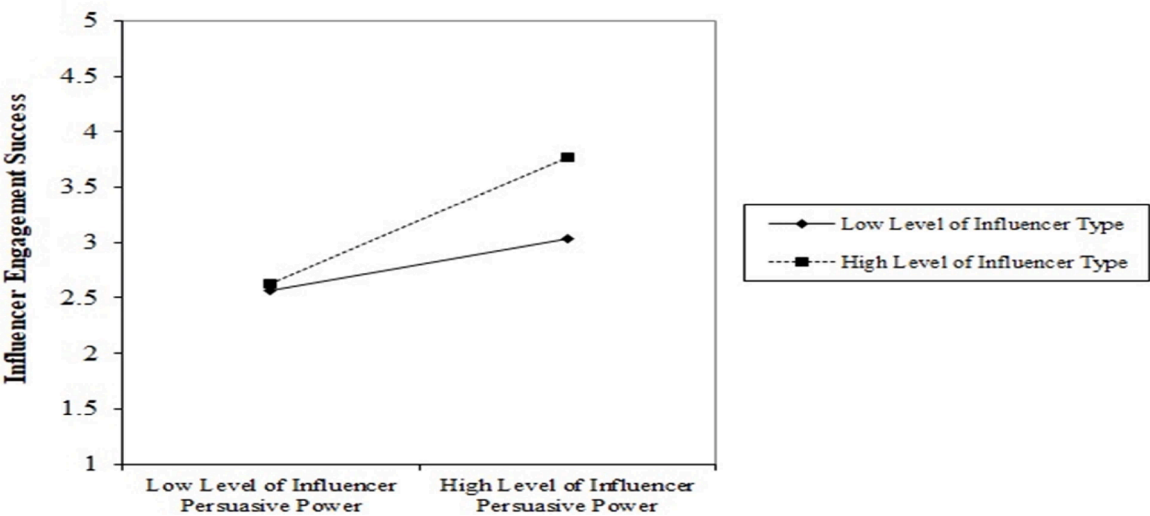


Fig. 4. Moderation effect of influencer type on the relationship between influencer persuasive power and influencer engagement success.

judgments based on the strength and relevance of the messages, leading to higher engagement. This finding indicates that posts imbued with persuasive elements encourage more profound interaction and analysis, corroborating the systematic aspect of the HSM theory. Influencer type can be a heuristic cue depending on the follower count size. For example, a high follower count may serve as a cue for popularity and credibility, thus affecting the perceived success of the influencer. This finding aligns with the heuristic processing aspect of the HSM theory, indicating that followers often utilize such apparent cues to form judgments about the influencer's reliability and content value.

In line with the proposed third hypothesis, our analysis revealed that the number of followers is a reliable heuristic cue for a post's success in engagement. This approval aligns the conventional reliance on follower count as a symbol of credibility, popularity, and reach. Our fourth hypothesis findings resonate with the heuristic aspect of the HSM theory, where the type of influencer serves as a predominant cue that overshadows the impact of other individual appearance characteristics. This finding suggests that, as follower count increases, the heuristic processing becomes more reliant on the influencer type, potentially diminishing the influence of other personal attributes on user engagement and post success. The support for our fourth hypothesis aligns with the heuristic principles of the HSM theory, where the influencer type serves as a dominant heuristic cue, overshadowing the impact of other individual appearance characteristics. The diminishing effect of appearance characteristics with the increase in follower count accentuates the reliance on heuristic processing based on the influencer type, highlighting the shifting influence dynamics as the follower count grows.

Finally, as proposed in the fifth hypothesis, it was found that while the influencer type serves as a heuristic cue affecting the perceived credibility and attractiveness of the influencer, the persuasive power encapsulates systematic processing. Here, the audience engages deeply with the content, evaluating the strength and relevance of the persuasive elements in it. The interaction between influencer type and persuasive power can amplify the systematic processing, intensifying the impact on user engagement. Moreover, the robust *r*-squared value and significant goodness of fit indices values underscore the analytical rigor and the explanatory power of the proposed research model in capturing the multifold factors influencing influencer engagement success. These high goodness-of-fit metrics affirm the model's reliability, substantiating the implications drawn from the supported and refuted hypotheses.

7. Contributions, limitations, and future research

7.1. Theoretical contribution

This study has several theoretical contributions, extending the extant literature. As the first contribution, as opposed to what extant literature suggests on HSM, our research findings showed no significant positive association between young female influencers and higher engagement. This misalignment with the heuristic cues extends the HSM theory, representing a contrary finding to what is proposed in HSM as the importance of heuristic cues. In this specific case, not all the heuristics would be as effective as thought earlier. Moreover, the validation of H5 intertwines both heuristic and systematic components of the HSM theory. The elevation of a post's persuasiveness with the increase in follower count illustrates how the influencer type acts as a heuristic cue, enhancing the systematic processing of the persuasive elements within the content. This synergy amplifies user engagement, reinforcing the relationship between heuristic cues and systematic processing in influencing perceptions and engagements. This rearticulated connection between the revised hypotheses and the Heuristic-Systematic Model magnifies the importance of understanding the intricate interplay between heuristic cues and systematic processing and extends the extant literature on digital influence.

As another theoretical contribution, this study extends the social

influencer literature by conceptualizing and validating two constructs of the persuasive power of an influencer in social media and influencer engagement success in social media. This exploration into digital influence, viewed through the lens of the HSM theory, unveils complex interdependencies between heuristic cues and systematic processing. The nuanced understanding obtained from the interrelation between individual appearance characteristics, influencer type, persuasive power, and their consequent impacts on engagement success provides valuable insights into the evolving nature of digital interactions and influence extant literature. The diverging results prompt a deeper reflection on the existing constructs within the HSM framework, paving the way for future research to unravel the multifaceted dimensions of digital influence.

7.2. Practical implications

This research has several practical implications. One core implication is the augmentation of marketing strategies for businesses and brands. By understanding the correlation between an influencer's type, age, gender, and persuasive power, marketers can more effectively identify and collaborate with influencers, reaching broad audiences and engaging and persuading them. This alignment ensures that promotional content is not merely seen but has a tangible impact, fostering a behavioral change in the audience, ideally aligned with the marketing goals.

Furthermore, research findings about the relationship between follower count, engagement, and persuasive power can inform a firm's customer relationship management strategies, creating a tailored and symbiotic relationship with influencers. Firms can establish long-term partnerships with influencers aligned with their brand values and ethos, nurturing a mutually beneficial relationship that delivers sustained value over time. This strategic alliance can extend beyond mere transactional exchanges, evolving into collaborative product development, content creation, and brand-building initiatives. Moreover, for firms seeking to introduce new products or tap into novel markets, influencers with demonstrable persuasive power can serve as strategic conduits to mitigate entry barriers, facilitate smoother market penetration, and expedite brand acceptance among potential consumers. Moreover, this research is critical in molding content strategies for influencers, enabling them to understand and utilize the elements that bolster their persuasive power.

On a societal level, understanding the dynamics of influencer engagement and persuasion brings to light influencer culture's psychological and social implications. Influencer endorsements and lifestyle portrayals can significantly sway consumers, especially younger demographics. The research could provide insights into developing educational initiatives to foster media literacy among the public. Knowledge about the mechanics of influencer persuasion could be utilized to educate social media users about critical engagement with online content, enabling them to navigate digital spaces with an informed and discerning perspective, potentially mitigating undue influence, and fostering healthier social media consumption patterns.

7.3. Limitations and future research

As with all research studies, the present study has its limitations. Initially, the research confines its data collection to a single social media platform, specifically Instagram, posing a potential for platform-specific bias in its findings. Consequently, future investigations must extend this research across various social media platforms to validate the universal applicability of these findings. Furthermore, the data compiled in this research predominantly originates from social media content that garnered attention from its users, presenting a possible bias that could influence the widespread relevance of the results. Subsequent studies may consider diversifying the data by exploring various influencer groups and examining an expansive array of content across the social

media spectrum.

While secondary data unveils unique perspectives, it curtails the researchers' capability to explore various attributes contributing to successful influencer engagement on social media. Studies might contemplate employing alternative research methodologies to navigate this substantial limitation. Moreover, our study is limited by the reliance on publicly available data, primarily including influencers who disclose their age and gender. Additionally, the dataset contains limited information on non-binary gender identities, restricting our ability to explore gender diversity among influencers entirely. These limitations are acknowledged to inform the interpretation of our findings and the scope of our analysis. Future research should aim to include a more diverse representation of gender identities and explore platforms where influencers may not publicly disclose demographic information to provide a more comprehensive understanding of influencer dynamics.

Additionally, this research did not scrutinize the nature of content that captivated social media users and catalyzed heightened engagement levels for influencers, presenting a potential avenue for future research to explore. Lastly, this study did not control the integrity of the content analyzed. Including potentially deceptive fake posts, which went unchecked due to the absence of monitoring mechanisms, poses a notable limitation that warrants attention in subsequent research endeavors.

8. Conclusion

In this study, we explore the factors that determine the success of social media influencers. Successful influencers were those whose advertising posts resulted in increasing user engagement. We contribute to the literature by simultaneously accounting for individual and persuasive appearance characteristics and developing a new model based on the HSM framework. We further investigated the relationship by including the effect of influencer type in our model.

Apart from the theoretical contributions, we offer a more accurate model of predicting influencer success with valuable practical implications. These implications are beneficial as consumers are more likely to change their behavior through influencer recommendations than traditional advertisements. As producers switch from traditional marketing means to novel approaches such as influencer marketing, it becomes increasingly essential to identify highly influential individuals capable of promoting their brands more effectively and spreading their message further. Our model of predicting influencer success can thus benefit marketing organizations and brands in choosing the right influencer.

CRedit authorship contribution statement

Pouyan Eslami: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Mahdi Najafabadi:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Amir Gharehgozli:** Data curation, Investigation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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